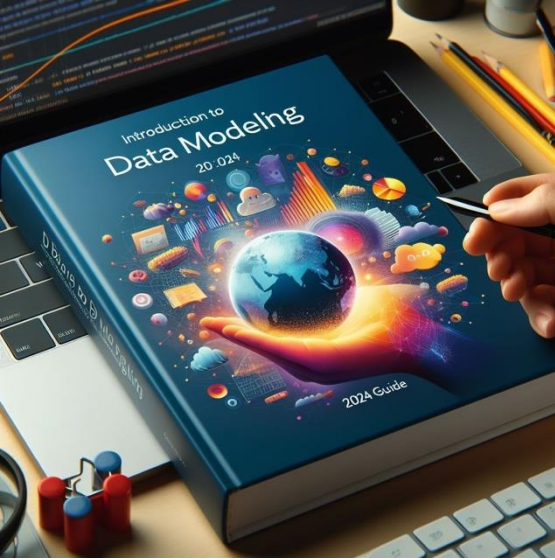




CSCE 411/811

Introduction to Data Modeling

Dr. Qiuming Yao

Fall 2025



College of Engineering

SCHOOL OF COMPUTING

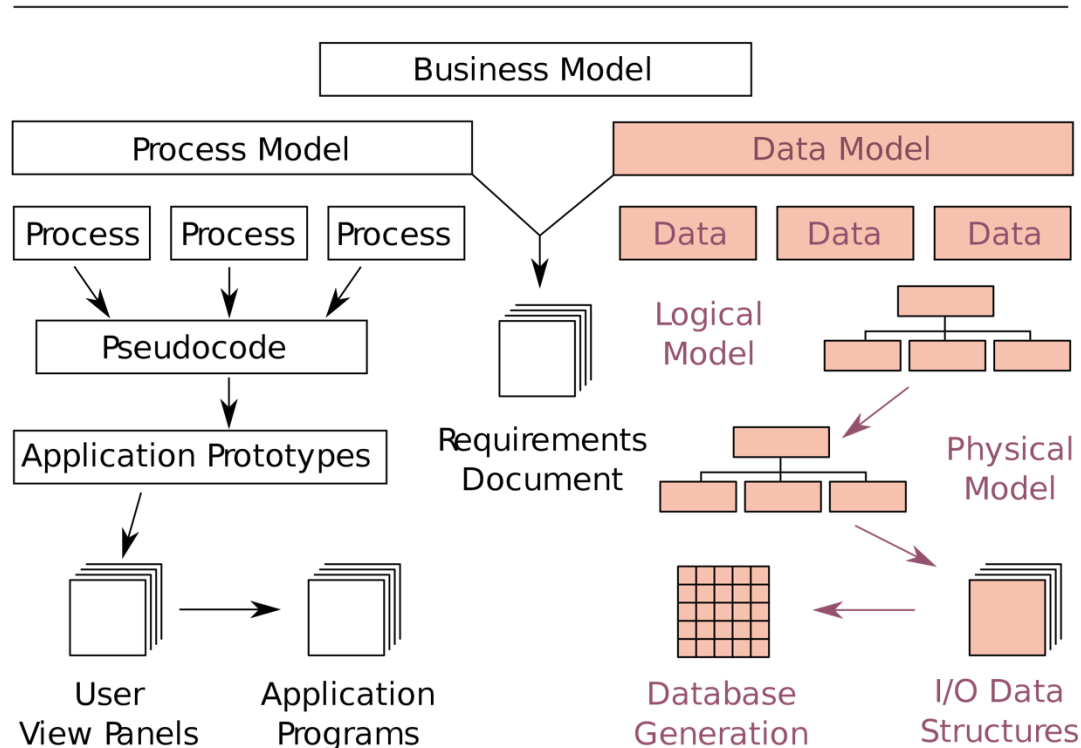
Course goals

- At the end of this course, you will be able to:
 - (1) Learn major concepts of data modeling
 - (2) Design data models in different processes
 - (3) Use different types of database design strategies
 - (4) Implement methods in data representation, modeling, and visualization

Data Modeling

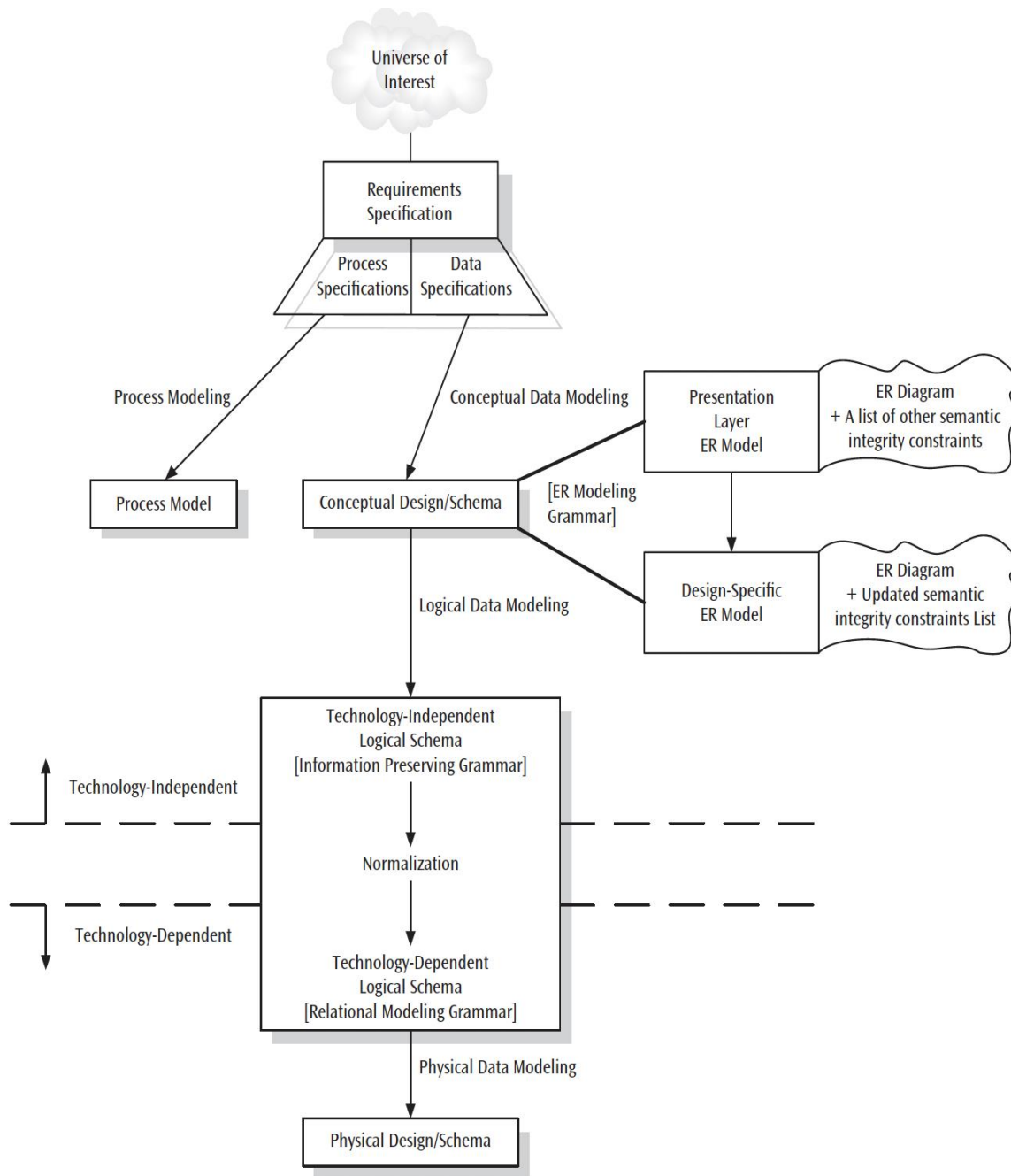
- Data Modeling supports your businesses

Business Model Integration



Data Modeling

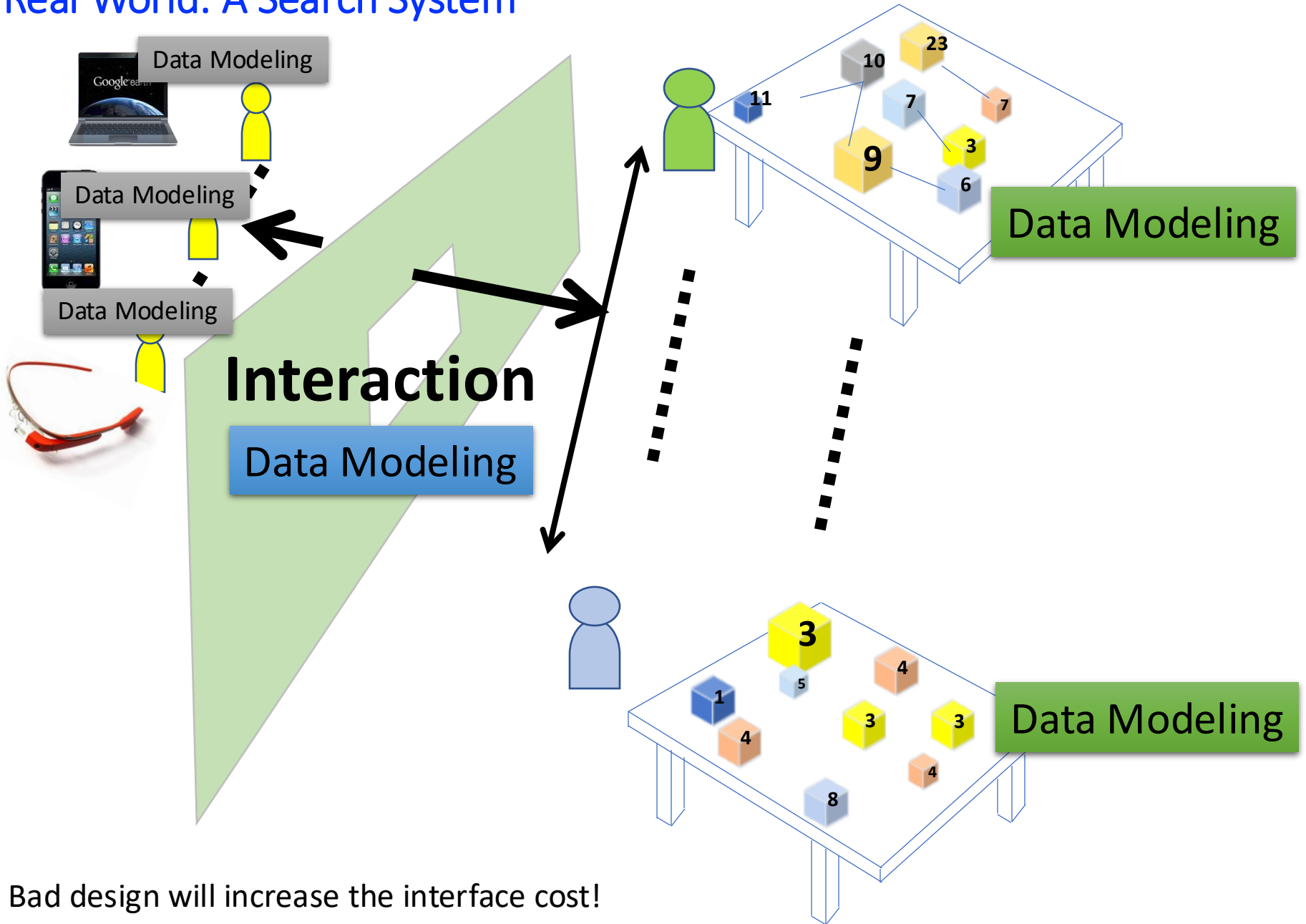
- Database management system (DBMS)
 - Data collection/preprocessing
 - Data representation
 - Data re-structuring
- Application processes
 - Data visualization
 - Data analysis
- Data communication
 - Data transmission
 - Data security
 - Data I/O processes



Phases of data modeling

PHASE 1 Conceptual model	PHASE 2 Logical model	PHASE 3 Physical model
Assigns properties for each component. Identifies data relationships (often business data relationships). Important to work with business-side groups to determine data relationships.	Creates unique data identifiers and determines the sources of data. Provides explicit identification of data sources. Provides the data architecture framework that will guide the physical model.	Dictates the structure of the actual database implementation. Allows database administrators to move forward with planning. Often best to work with database management tools already available to your organization, as adopting new vendors can be expensive.

Real World: A Search System



Bad design will increase the interface cost!

Example

- Bad design increases your business cost!

Table 1.2 A badly-designed relational database table.

Movie:

<i>movieNr</i>	<i>movieTitle</i>	<i>releaseYr</i>	<i>director</i>	<i>star</i>
1	Cosmology	2006	Lee Lafferty	?
2	Kung Fu Hustle	2004	Stephen Chow	Stephen Chow
3	The Secret Garden	1987	Alan Grint	Gennie James
3	The Secret Garden	1987	Alan Grint	Barret Oliver
4	The Secret Garden	1993	Agnieszka Holland	Kate Maberly
4	The Secret Garden	1993	Agnieszka Holland	Heydon Prowse
5	The DaVinci Code	2006	Ron Howard	Tom Hanks
5	The DaVinci Code	2006	Ron Howard	Ian McKellen
5	The DaVinci Code	2006	Ron Howard	Audrey Tautou

Example

- Student class enrollment system
- What are the users?
- What are typical user cases?
- What are the essential data?
- How to organize the data to support user cases?
- What shall the data representation and relationships?
- What is the DBMS needed?

Covered Topics

- Introduction
- Concerns of design:
efficiency? time & space?
user experience? visualization?
- Diagram and modeling
ER, EER, UML, ORM
- Structured data vs. Unstructured data

Convergent design

- When all stories converge to something, that's your system.
- Diagrams are useful and important to assist or lead data modeling and system design.